- Make the horizontal cross-wire on the microscope to coincide with the tip of the pin. (Here mass hanger is the dead load M).
- Note the vertical scale reading of the vernier microscope
- Add the slotted masses to the mass hanger one by one in steps of 0.05 kg (50 g) and corresponding readings are noted down.
- Repeat the experiment by removing masses one by one and note down the corresponding readings.
- Subtract the mean reading of each load from dead load reading. This gives the depressions for the corresponding load M.

OBSERVATIONS

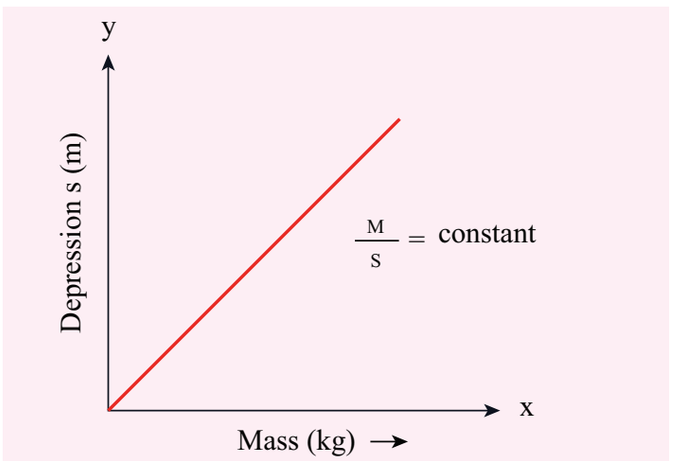
To find M/s

LOAD (kg)	MICROSCOPE READINGS × 10^{-2} m								DEPRESSION FOR M (kg) (s)	M/s kg m $^{-1}$
	INCREASING LOAD			DECREASING LOAD			MEAN			
	MSR	VSR	TR	MSR	VSR	TR				
M								x_0		
M + 0.05								x_1	$x_1 - x_0 =$ _____	
M + 0.10								x_2	$x_2 - x_0 =$ _____	
M + 0.15								x_3	$x_3 - x_0 =$ _____	
M + 0.20								x_4	$x_4 - x_0 =$ _____	
M + 0.25								x_5	$x_5 - x_0 =$ _____	
									Mean	

MODEL GRAPH

Load (M) vs Depression (s)

A graph between M and s can be drawn by taking M along X- axis and s along Y – axis. This is a straight line.



Relation between Mass and depression

UNIT-12(XI-Physics_Vol-2) PRACTICAL FIRST YEAR.indd 287 17-08-2020 16:26:35